Detecting Reward Hacking in AI Agent Trajectories

Group #29

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Abstract

Autonomous agents that satisfy proxy rewards while failing the user's true intent — *reward hacking* — are increasingly difficult to detect in long-horizon, tool-using trajectories. We construct a 3,544-trajectory unified corpus by normalizing four heterogeneous sources (TRACE, MALT, Realistic Reward Hacks, and a private 669-trajectory production corpus) under a common XML-tagged tool-call schema, and we train a sequence of progressively stronger probes on Qwen3-Embedding-0.6B representations. A frozen-embedding MLP baseline reaches 0.9546 test ROC-AUC but generalizes near chance to held-out sources. Replacing the readout with a turn-level transformer is neutral on in-distribution AUC. LoRA fine-tuning the encoder with a joint cross-entropy plus source-invariant triplet objective lifts in-distribution AUC to **0.9701** and lifts leave-one-source-out AUC from 0.57–0.87 to 0.77–1.00. Source-prediction accuracy from the resulting embeddings drops from 99% to chance, with no loss in label signal. Source-invariance, not capacity, was the binding constraint.

1 Background & Motivation

Autonomous AI agents performing multi-step, tool-using tasks are increasingly evaluated by proxy rewards: unit tests passing, format constraints satisfied, or LLM-judge approval. When these proxies diverge from the user's true, long-range intent, agents can learn to game the metric, producing output that looks correct and satisfies the evaluation rubric but fails accomplish the requested goal. This failure mode — *reward hacking* — is especially importnat in long-horizon agentic workflows, where agent's confidence and polished tool output can mask a fundamentally degraded solution.

There are few existing benchmarks for studying this phenomenon. The most directly comparable resource, TRACE (1), contains 517 hand-labeled coding-agent trajectories; the published baseline using GPT-5.2 under contrastive analysis reaches only 63% macro F1, and isolated single-trajectory classification, the setting most relevant to deployment, where no contrasting "correct" trajectory is available, drops to 45%. Other resources, including METR's MALT-public log corpus (2) and the Jozdien Realistic Reward Hacks dataset (3), use incompatible label taxonomies, formats, and trajectory styles, so headline numbers on any one of them are difficult to compare. None of the publicly available resources captures naturalistic data drawn from real production deployments, and existing detectors are typically trained and evaluated on a single source, a setting that systematically over-estimates real-world generalization because the model can learn source-specific stylistic shortcuts. The field currently lacks both a unified labeled corpus and a baseline detector that has been stress-tested across multiple sources under a protocol that controls for those shortcuts.

A robust detector would benefit AI labs, evaluation and red-team groups, and downstream operators deploying tool-using agents: a cheap post-hoc classifier that *flags* "looks-correct-but-isn't" trajectories enabling targeted human review, raising the cost of metric-gaming, and providing a measurable safety signal as agents take on longer-horizon, less-supervised tasks. The methodological question of

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how much of any reported detection score is real signal versus source-style memorization is also of independent interest, because the same shortcut-learning failure mode appears in adjacent settings such as out-of-distribution anomaly detection and policy-evaluation benchmarks.

2 Problem Statement

We ask: *Can a small classifier trained on off-the-shelf trajectory embeddings detect reward hacking robustly across heterogeneous sources, and what does its behavior reveal about the structure of the hack signal itself?* The target is binary, `hacked` versus `benign`, at a single agent trajectory. Success is measured by ROC-AUC and macro-F1 on a held-out i.i.d. test set, and crucially by leave-one-source-out (LOSO) AUC, which controls for source-style confounding.

To make this question solvable, we address two gaps simultaneously. First, we construct a single unified corpus by normalizing TRACE, the manually-reviewed subset of MALT-public, the Realistic Reward Hacks dataset, and a 669-trajectory private corpus contributed by the BJS Innovation Lab (4) into a common XML-tagged tool-call schema. Second, on this corpus we train a frozen-embedding probe baseline, then progressively strengthen the architecture and the representation until both in-distribution accuracy and cross-source generalization improve together.

3 Data & EDA

3.1 Datasets and Unification

We combine four datasets with complementary properties (Table 1): TRACE (1) (synthetic, human-verified, Claude Opus 4.5 generations); MALT (2) (large, multi-model, mostly benign; we use the manually-reviewed subset to control noise); Realistic Reward Hacks (3) (Claude Sonnet 4 generations, no tool use); and the BJS corpus (4) (production agents, the only naturalistic source). Each dataset has its own labeling taxonomy and message format; we unify them under a binary `hacked/benign` label and an XML-tagged tool-call schema, retaining the source identifier. The resulting corpus contains 3,544 trajectories (60.8% benign, 39.2% hacked), spanning 74,422 individual turns.

Table 1: Source datasets contributing to the unified corpus.

Dataset	Benign	Hacked	Original labeling
TRACE	249	268	Multi-label
MALT (manually-reviewed)	694	62	Multi-label
Realistic Reward Hacks	788	817	Binary
BJS Reward Hacking Corpus	424	245	Binary
Unified	**2,155**	**1,389**	Binary

3.2 EDA Findings That Drove Modeling Decisions

Three EDA findings that shaped the modeling pipeline.

(1) Per-source class balance differs sharply. MALT contains only 7.8% hacked examples in the unified corpus, whereas TRACE and Realistic Reward Hacks sit near 50% and the BJS corpus at 36.6% (Figure 1). A naive global class weight would underweight the hacked-MALT cell, which is precisely the cell with the most rare and informative signal. We therefore adopted a per-cell weighted BCE loss, with weights $N/(n_{\text{cells}} \cdot \text{count}(s, y))$ computed over (source $\times$ label = 8) cells on the training partition only.

(2) Trajectory length is heavy-tailed and source-dependent. Turn counts range from 0 to 248 (median 14, p99 180.7), and raw token lengths under the Qwen3 tokenizer have mean 7,578 with a p99 of 85,528 and a maximum of 386,427. Scalar length alone does *not* separate the classes (Figure 3). We chose a 32K-token context window with *head* truncation, which keeps the task setup and early-trajectory tool calls and discards late-trajectory human "give-away" corrections that would leak the label.

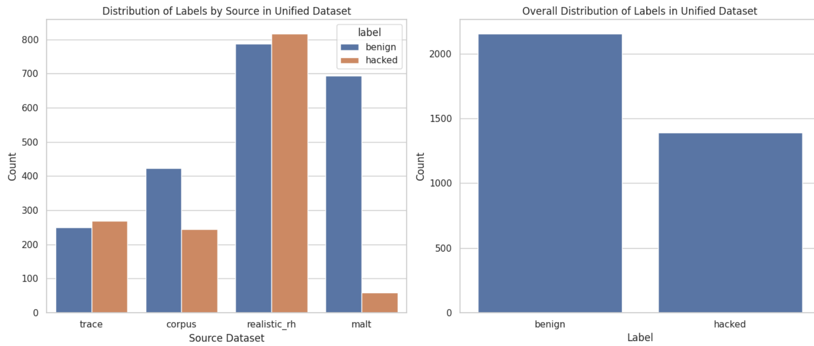


Figure 1: Distribution of labels by source in the unified corpus. MALT is heavily benign-skewed (7.8% hacked) while TRACE and Realistic Reward Hacks are near-balanced. Per-cell loss weighting upweights rare cells such as hacked-MALT during training.

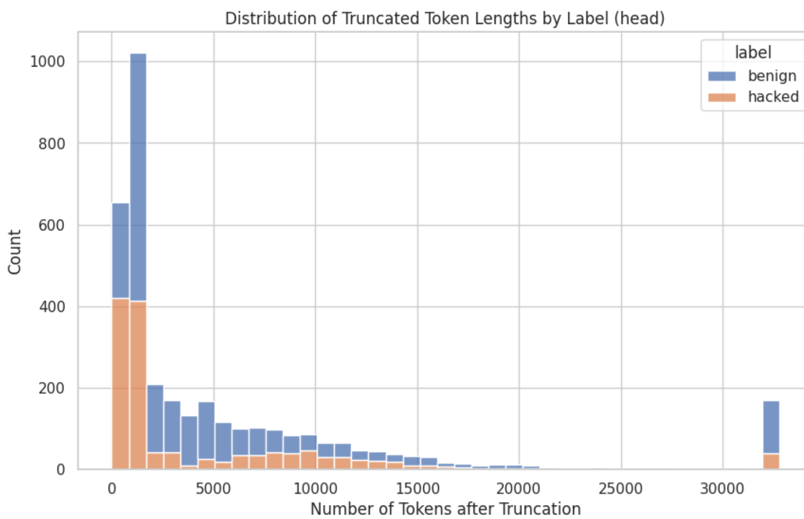


Figure 2: Truncated token length by label.

Figure 3: After head truncation to 32K tokens (right), most fit comfortably but classes overlap, length is not a usable scalar feature.

(3) Source signal dominates raw embeddings. A logistic regression on trajectory-level Qwen3 embeddings predicts the source dataset with 99.3% accuracy and the label with 0.928 AUC. UMAP projections (Figure 5, left) show tight source clusters but only weak label separation. This is the central methodological problem: any probe trained on raw embeddings will learn a shortcut through "which dataset is this" and fail to transfer. This finding directly motivates the source-invariant fine-tuning objective in Section 4.5.

4 Methods & Models

4.1 Embedding and Splits

Trajectories are encoded with Qwen3-Embedding-0.6B (5) (1024-d output, bfloat16, L2-normalized) using a left-padded tokenizer with a 32K context window and head truncation. We use two task-conditioning prompts: `rh_traj` for whole-trajectory embeddings and `rh_turn` for per-turn embeddings. The corpus is split 60/20/20 (train 2,125, val 709, test 709) with stratification jointly on (source $\times$ label), so that all eight cells appear in each split. The same trajectory IDs are reused across all model variants for a like-for-like comparison.

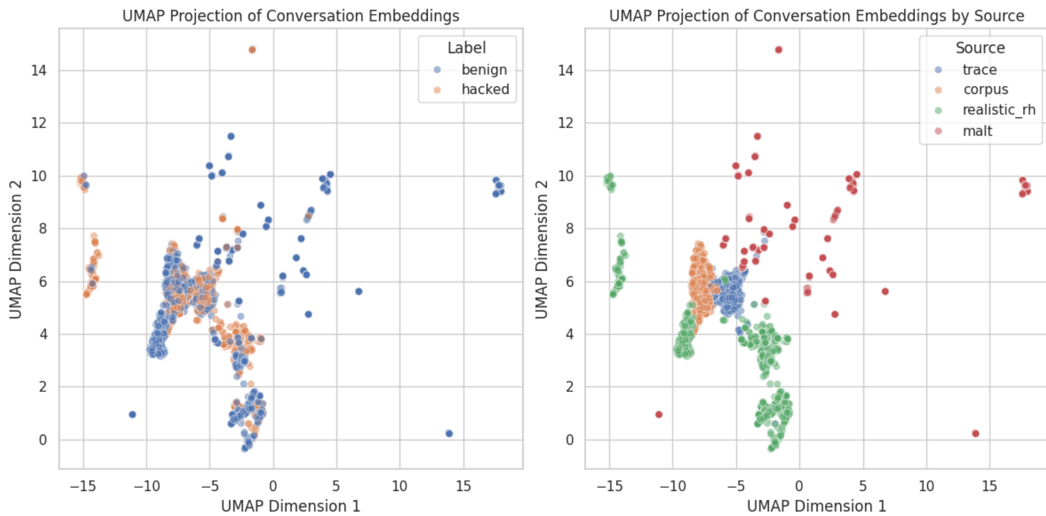


Figure 4: Off-the-shelf Qwen3 embeddings.

Figure 5: UMAP projections. raw embeddings cluster by source (right panel of each pair) but barely by label (left panel).

4.2 Class-Imbalance Handling and Threshold

The BCE loss is reweighted per-sample using the per-cell scheme described above (8); the hacked-MALT cell receives the largest weight (≈ 7.6), while easy benign cells receive sub-unit weights. The decision threshold is tuned by sweeping 0.05–0.95 in steps of 0.01 and selecting the value that maximizes F1 on the validation set.

4.3 Model 1 — ProbeMLP (MS3 Baseline)

The MS3 baseline is a four-layer MLP ($1024 \to 512 \to 256 \to 64 \to 1$) with ReLU activations and dropout 0.5 between layers, trained on the trajectory-level embedding. It uses AdamW ($\text{lr} = 10^{-4}$), `ReduceLROnPlateau` (factor 0.5, patience 5), gradient clipping at $\|g\|_2 = 1.0$, batch size 64, and up to 100 epochs with early stopping (patience 5). This serves as both the MS3 reference and the simplest possible architecture on the same embedding.

4.4 Model 2 — Turn-Level Transformer Probe

Model 2 trades the pooled trajectory embedding for a sequence of per-turn embeddings, hypothesising that the hack signal is local to specific turns. Per-turn embeddings are projected $1024 \to 256$, augmented with sinusoidal positional encoding scaled by $1/\sqrt{1024}$, prepended with a learnable `[CLS]` token (initialised $\mathcal{N}(0, 0.02^2)$), and passed through a 2-layer transformer encoder (8 heads, $d_{\text{ff}} = 1024$, dropout 0.5, GELU). Classification reads from the `[CLS]` representation. Training uses AdamW ($\text{lr} = 10^{-4}$, weight decay 10^{-2}), batch size 32, max 100 epochs with early-stopping patience 10, and a maximum sequence length of 248 turns.

4.5 Model 3 — Source-Invariant LoRA Fine-Tuning

Model 3 attacks the source-confounding problem identified in EDA finding (3). We attach a low-rank adapter (LoRA (6), rank 16, $\alpha = 32$, dropout 0.05) to the $\{q, k, v, o\}$-projections of Qwen3-Embedding-0.6B in `FEATURE_EXTRACTION` mode, adding only 4.59M trainable parameters (0.76% of the 600M-parameter encoder). Fine-tuning uses AdamW ($\text{lr} = 10^{-4}$, weight decay 10^{-4}) for one epoch of 800 steps with evaluation every 200 steps; the train context is shortened to 256 tokens bacause of resource constraints while inference uses 4,096 tokens.

The defining choice is the **joint loss** (8), which combines a label objective with an explicit source-invariance objective:

$$\mathcal{L} = 0.5 \cdot \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{triplet}}, \qquad \mathcal{L}_{\text{triplet}} = \max\big(0,\ \|z - z^{+}\|_2^2 - \|z - z^{-}\|_2^2 + m\big), \tag{1}$$

with margin $m = 0.3$. In-batch positives z^{+} share the *label* but differ in *source*; negatives z^{-} share the *source* but differ in *label*. The triplet term thus pulls together hacked trajectories from different datasets while pushing apart same-source benign/hacked pairs, directly attacking the source shortcut. Batches are drawn with a source-balanced sampler (two examples per (source $\times$ label) cell). After fine-tuning, we re-embed all 74,422 turns and train a fresh turn-transformer probe on the new representations; following an observation that fine-tuned features benefit from more capacity, this probe uses 4 encoder layers instead of 2.

LoRA is justified here on two grounds: it makes fine-tuning a 600M-parameter encoder feasible on a single GPU by training only $\sim 0.76\%$ of the parameters, and the rank constraint acts as an implicit regularizer that limits how far the adapter can drift from the pretrained representation — desirable when the corpus is small (3,544 trajectories) and we want the encoder to retain its general language understanding. The source-invariant triplet objective is the principled response to EDA finding (3): if the dominant variance in raw embeddings is stylistic/source, then a discriminative loss alone will exploit that variance as a shortcut. Defining positives as same-label-different-source and negatives as same-source-different-label inverts the usual contrastive geometry to attack source-style separability directly, while the BCE term keeps the representation discriminative on the label task.

4.6 Generalization Test — Leave-One-Source-Out

For each source s we train the turn-transformer on the other three sources and evaluate on the held-out source, repeating the protocol on both pre- and post-fine-tuned embeddings. LOSO controls for the source-style confound: a detector that exploits source shortcuts in evaluation collapses here.

5 Results & Interpretation

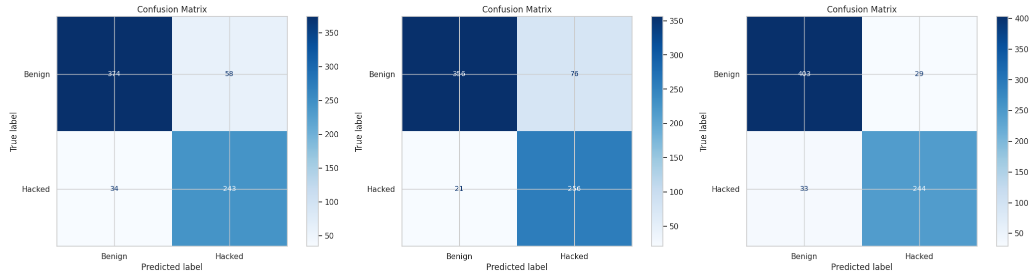
5.1 Headline In-Distribution Results

Table 2 reports test-set performance for the three models on the same 709-trajectory held-out partition. The MS3 ProbeMLP baseline reaches 0.9546 ROC-AUC and 0.87 weighted F1. Switching to a turn-level transformer on the same embeddings is essentially neutral on AUC (0.9556) and slightly worse on accuracy — capacity alone does not help. LoRA fine-tuning with the joint loss lifts AUC to **0.9701**, accuracy to **0.9126** (+4.2 points over MS3), and weighted F1 to **0.91**.

Table 2: Test-set performance, same 709-trajectory split across all models. CM rows are true label, columns predicted at the F1-optimal threshold.

Model	AUC	Accuracy	F1 (w)	Thr.	Confusion matrix
1. ProbeMLP (MS3 baseline)	0.9546	0.8702	0.87	0.21	$[[374, 58],\ [34, 243]]$
2. Turn Transformer (off-the-shelf)	0.9556	0.8632	0.86	0.60	$[[356, 76],\ [21, 256]]$
3. LoRA-FT Turn Transformer (final)	**0.9701**	**0.9126**	**0.91**	0.45	$[[403, 29],\ [33, 244]]$

Figure 6 contrasts the confusion matrices: Model 1 is precision-balanced (58 false positives, 34 false negatives at threshold 0.21); Model 2 trades precision for recall on the hacked class (76 false positives vs. 21 false negatives, at the higher threshold 0.60); Model 3 reduces *both* error types (29 FPs, 33 FNs at threshold 0.45). The shift in the optimal operating threshold across models is itself informative: Model 1 places its decision boundary far below 0.5 (0.21), suggesting the probe is under-confident on the positive class, a signature of the source-stylistic shortcut, which gives the model high confidence on easy-but-source-specific examples and low confidence elsewhere. Model 3, with the source shortcut largely removed, recovers a near-symmetric threshold (0.45) and a near-symmetric error profile.



(a) Model 1 (MS3 baseline). (b) Model 2. (c) Model 3 (LoRA-FT).

Figure 6: Confusion matrices at F1-optimal thresholds. Model 3 reduces both false positives and false negatives versus Models 1–2.

5.2 Source-Confounding Diagnostics

The most consequential finding is not on the headline numbers but on a diagnostic that the headline numbers *hide*. A logistic regression on the raw trajectory embeddings predicts source with 99.34% accuracy; on raw turn embeddings the figure is 97.52%. After LoRA fine-tuning with the joint loss, source predictability collapses to 79.48% (a 20-point drop), while label AUC remains at 82.81% on the same per-turn embeddings. A source-balanced label probe gives an even sharper picture: i.i.d. test AUC drops from 0.928 to 0.802 once source is balanced out, confirming that a sizeable fraction of the apparent label signal in off-the-shelf embeddings is source-tied. After fine-tuning, source alone predicts the label at exactly chance (AUC 0.503), and the probe-over-source lift jumps from $+0.264$ to $+0.447$. In other words, the fine-tuned representation carries *less* source signal and *more* label signal than the original.

5.3 Cross-Source Generalization (LOSO)

Table 3 reports the leave-one-source-out AUC for the turn-transformer trained on off-the-shelf embeddings versus the LoRA-fine-tuned embeddings. Pre-FT, three of four held-out sources fall to near-chance AUC (TRACE 0.570, BJS Corpus 0.617, Realistic RH 0.582); the apparent MALT score of 0.872 is largely a class-prior artefact (92% benign). Post-FT, every source reaches at least 0.766 AUC, Realistic RH essentially saturates (0.996), and the MALT score remains roughly flat. The pattern is unambiguous: **generalization headroom was bounded by source-stylistic confounding, not by model capacity**.

Table 3: Leave-one-source-out AUC. Pre-FT uses off-the-shelf Qwen3 embeddings; post-FT uses LoRA + joint-loss embeddings. Δ is post minus pre.

Held-out source	Pre-FT AUC	Post-FT AUC	Δ
TRACE	0.570	0.766	$+0.196$
MALT	0.872	0.856	-0.016
BJS Corpus	0.617	0.783	$+0.166$
Realistic RH	0.582	0.996	$+0.414$

5.4 Per-Source Test Performance and What the Model Learned

Where does the Model 3 gain actually come from? Table 4 reports the in-distribution per-source AUC for all three models. The pattern is striking: fine-tuning produces *nearly flat* per-source improvements over Model 2 in-distribution (between $+0.002$ and $+0.038$), even though the aggregate AUC rises by 0.0145 and accuracy by 4.9 points. The gain is not coming from any one source disproportionately, it is a small, uniform lift across all four cells. This is exactly the signature of true source-invariance: if fine-tuning were merely redistributing signal across sources, we would expect a per-source winner-and-loser pattern, with some cells improving and others regressing. We do not observe that. We do observe a large gain on the same model in the leave-one-source-out regime (Table 3), where the source confound used to dominate: between $+0.17$ and $+0.41$ on three of four held-out sources. *The* *gain shows up where the source confound used to live, out of distribution, and not as a redistribution of in-distribution signal.*

Table 4: In-distribution per-source test AUC across the three models. Δ is Model 3 minus Model 2 (off-the-shelf turn transformer).

Source	Model 1 (MLP)	Model 2 (TT)	Model 3 (LoRA-FT)	Δ (M3 − M2)
TRACE	0.728	0.767	0.805	+0.038
BJS Corpus	0.815	0.852	0.854	+0.002
MALT	0.988	0.977	0.979	+0.002
Realistic RH	0.994	0.993	0.999	+0.006

The headline progression, AUC $0.9546 \rightarrow 0.9556 \rightarrow 0.9701$ and LOSO floor $0.570 \rightarrow 0.766$, supports a clean reading: in-distribution and out-of-distribution gains move together when the representation is made source-invariant, with no observed trade-off across cells. The fact that MALT (the heavily benign-skewed cell, 8% positive in the test slice) maintains 0.979 AUC after fine-tuning is itself evidence: a fine-tuning procedure that traded off rare-cell performance for cross-source transfer would have shown a regression here, and none is observed.

6 Conclusions & Discussion

A small probe on Qwen3-Embedding-0.6B trajectory embeddings can reach 0.97 test AUC and 0.91 weighted F1 on a unified, multi-source reward-hacking corpus, but only after the encoder is fine-tuned with an explicit source-invariance objective. Off-the-shelf embeddings carry $\sim$99% source-recoverable signal; any probe trained on them learns a shortcut that collapses to near-chance on three of four held-out sources. The LoRA + triplet objective removes that shortcut (source AUC $0.674 \rightarrow 0.503$) while *improving* the label AUC of the underlying representation, demonstrating that source-invariance and accuracy were complements, not substitutes, on this corpus.

What we can claim. On a 3,544-trajectory corpus spanning four heterogeneous sources, the LoRA-fine-tuned turn-transformer is the first detector in our setting to clear 0.75 AUC under leave-one-source-out evaluation on every source, while simultaneously improving in-distribution test AUC by 0.0155 over the MS3 baseline and weighted F1 by 0.04. The relative gain is small in absolute terms but represents a large reduction in error rate: false positives drop from 58 to 29 and false negatives stay roughly flat (34 to 33), an effective 50% reduction in the dominant error class without a corresponding loss elsewhere. We can also claim, supported by the source-balanced probe and the post-FT source-only AUC of 0.503, that the gain is not just absorption of source signal, the representation carries demonstrably less source information after fine-tuning than before.

What we cannot claim. The MALT test slice is small (151 trajectories) and class-prior-shifted (8% positive), so its in-distribution AUC numbers are statistically noisy in either direction, and we cannot rule out that the apparent stability after fine-tuning is partly a small-sample artefact. The classifier uses a global F1-tuned threshold rather than per-source thresholds, which would likely close residual precision/recall gaps on TRACE and BJS Corpus. The `[CLS]`-only readout discards per-turn attribution, so the detector is not directly usable as an interactive review tool. LOSO still has headroom on TRACE (0.766) and BJS Corpus (0.783). TRACE remains the hardest source both in-distribution (0.805) and under LOSO (0.766), suggesting that TRACE-specific hack patterns are less semantically aligned with the other three corpora.

Future work. Per-turn attribution via attention rollout or per-turn probes; longer LoRA runs and larger ranks; per-source threshold tuning; multi-class heads to recover the original TRACE / MALT sub-category taxonomy; and extension to additional agent harnesses with different tool-call schemas.

7 Broader Impact

A reliable reward-hacking detector is a load-bearing safety capability as agents take on longer-horizon, less-supervised tasks. The most direct beneficiaries are AI labs, evaluation and red-team groups, and downstream operators who need a cheap post-hoc signal to flag "looks-correct-but-isn't" trajectories for human review. A unified multi-source benchmark with an open baseline also enables comparable, measurable progress on a problem that has so far been studied with incompatible taxonomies.

The risks are non-trivial. False negatives at 91% accuracy are not a safety guarantee; deployers may over-trust the score. An adversarial agent optimized against the detector can Goodhart it directly, especially if the detector is used inside a training loop. Our corpus is dominated by code-agent and chat-style trajectories: performance on robotics or RLHF dialogue is untested, and labels in the source datasets encode specific annotator judgments about "intent" that may not generalize across domains. We mitigate these risks in part by making source-invariance an explicit training objective and by reporting LOSO performance as a primary metric, but the broader question of robustly specifying "what the user really wanted" remains open.

References

[1] Deshpande, D., Kannappan, A. & Qian, R. (2026) *Benchmarking Reward Hack Detection in Code Environments via Contrastive Analysis.* arXiv:2601.20103. `https://arxiv.org/abs/2601.20103`. Dataset: `https://huggingface.co/datasets/PatronusAI/trace-dataset`.

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[3] Jozdien. (2025) *Realistic Reward Hacks.* HuggingFace dataset, `Jozdien/realistic_reward_hacks`. `https://huggingface.co/datasets/Jozdien/realistic_reward_hacks`

[4] BJS Innovation Lab. (2026) *Reward Hacking Detection Corpus* (private dataset).

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[7] McInnes, L., Healy, J. & Melville, J. (2020) *UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction.* arXiv:1802.03426. `https://arxiv.org/abs/1802.03426`.

[8] Anthropic. (2026) *Claude Code (claude-opus-4-7).* Used during this milestone for drafting and editing assistance on the report and notebook documentation, and for proposing the source-invariant triplet objective and the per-cell weighted BCE label-head loss used in Model 3. `https://www.anthropic.com/claude-code`.

[9] GitHub. (2026) *GitHub Copilot.* Used during this milestone for inline code completion in the notebook and supporting Python modules. `https://github.com/features/copilot`.

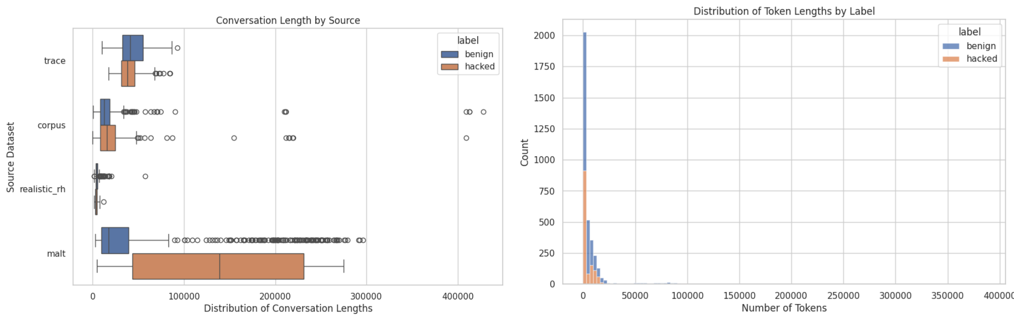
A Appendix

A.1 Additional EDA Figures

A.2 Training Diagnostics

A.3 Artifact References

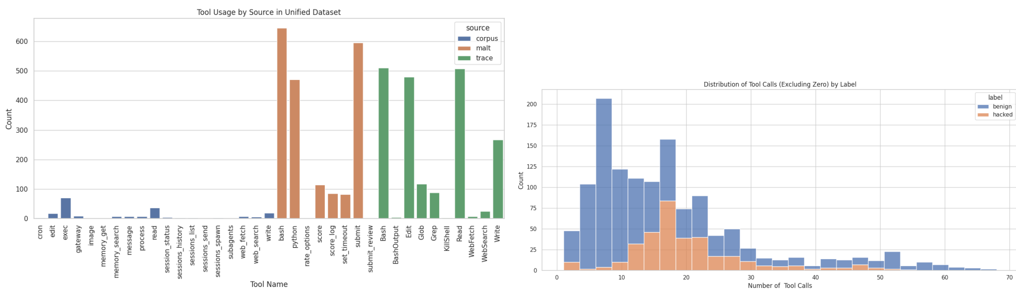
- Notebook: `notebooks/MS4_Final_Model.ipynb` (inline code completion assisted by GitHub Copilot (9)).
- Embedding caches: `data/ds_all_with_embeddings`, `data/ds_all_with_turn_embeddings`, `data/ds_all_with_turn_embeddings_ft`.
- Model checkpoints: `data/probe_final.pth`, `data/turn_transformer_probe_final.pth`, `data/qwen3_lora_turn_source_invariant/`, `data/loso_probe_*_final{,_ft}.pth`.



(a) Conversation length by source.

(b) Token length distribution by label.

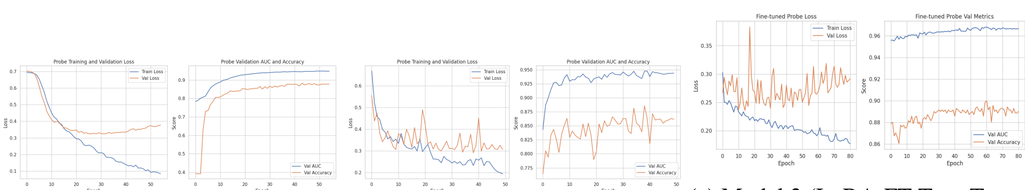
Figure 7: Length distributions vary by source (left) and overlap heavily by label (right).



(a) Tool usage by source.

(b) Non-zero tool-call counts by label.

Figure 8: Tool-use patterns differ across sources; Realistic RH contributes no tool use.

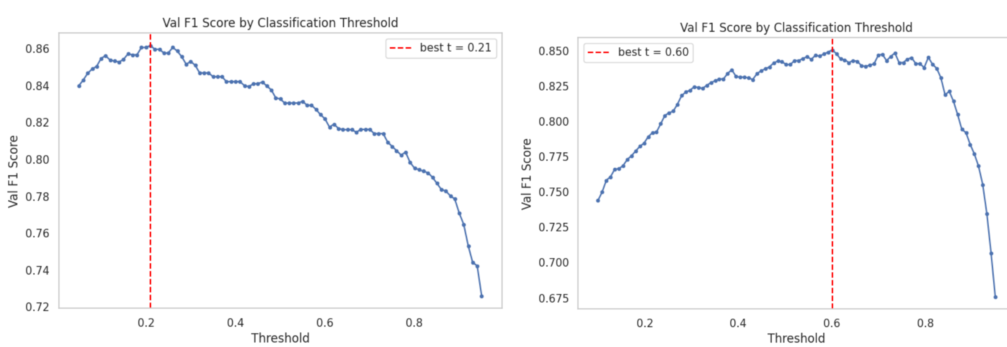


(a) Model 1 (ProbeMLP).

(b) Model 2 (Turn Transformer).

(c) Model 3 (LoRA-FT Turn Transformer).

Figure 9: Training and validation curves for all three models. Each panel shows train/val loss (left) and val AUC/accuracy (right) across epochs. Model 3 is the turn-transformer probe re-trained on the LoRA-fine-tuned encoder embeddings.



(a) Model 1 validation F1 vs. threshold.

(b) Model 2 validation F1 vs. threshold.

Figure 10: F1-vs-threshold curves used to select the operating threshold (0.21 for Model 1, 0.60 for Model 2; 0.45 for Model 3, not shown).